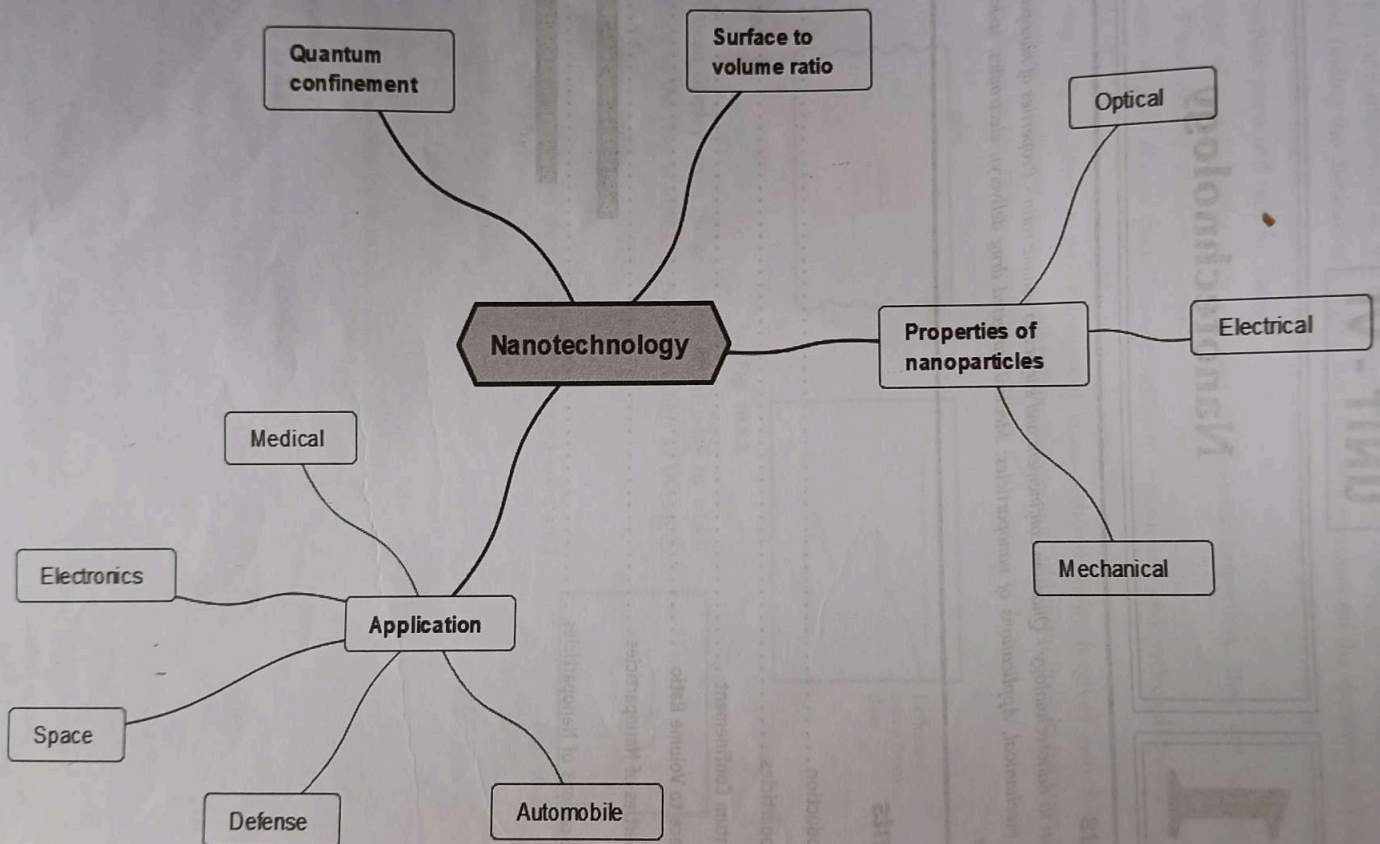


Mind Map - Nanotechnology



11.1 : Introduction

- The prefix 'nano' means a billionth (10^{-9}). The field of nanotechnology deals with various structures of matter having dimensions of the order of a billionth of a meter. These particles are called **nanoparticles**.
- Nanotechnology is based on the fact that particles which are smaller than about 100 nm give rise to new properties of nanostructures built from them.
- Particles which are smaller than the characteristic length for a particular phenomenon show different physical and chemical properties than the particles of larger sizes.
- For example, mechanical properties, optical properties, conductivity, melting point and reactivity have all been observed to change when particles become smaller than the characteristic length.
- Gold and silver nanoparticles were used in window glass panes to obtain a variety of beautiful colours.
- Nanotechnology has wide range of applications like producing lighter but stronger materials, constructing faster switches for computers, improving drug delivery to specific organs of the body etc.

11.2 : Nanoparticles

- The radii of atoms and most of the molecules are less than a nanometer.
- Nanoparticles are generally considered to have radius in the range of 1 nm to 100 nm which can have 25 to 10^6 atoms. A cluster of 1 nm radius has approximately 25 atoms.
- This definition of nanoparticles based on size does not distinguish between molecules and nanoparticles as many organic molecules contain more than 25 atoms.
- Nanoparticles can be more appropriately defined as an aggregate of atoms between 1 nm and 100 nm with dimensions less than the characteristic length of some physical phenomena.
- When particle size is less than the characteristic length of some physical phenomena, the particles show different properties.
- The nanoparticles show unique properties that change with their size.
- Classical mechanics is able to explain properties of bulk materials but is unable to explain properties of nanoparticles.
- Quantum mechanical principles have to be used to explain properties of the nanoparticles.
- The state of matter around the nanosize is known as **mesoscopic state**.

11.3 : Quantum Confinement

- Density of states is the possible electron quantum energy states between energies E and $E+dE$ per unit volume. The density of states for metals in three dimensions in a bulk material is given by

$$g(E) = \frac{8\sqrt{2} \pi m^{3/2}}{h^3} E^{1/2}$$

- Hence, the density of states increases with energy as shown in Fig. 11.3.1 (a). From the above equation, we can conclude that the density of states in a bulk material is proportional to $E^{1/2}$ and hence increases with energy.
- If the electrons are confined in one or more directions by reducing the dimensions of the material in those directions, the density of states changes due to quantization of energy.
- For a quantum well, which is a material reduced in one dimension to nano - scale, there is quantization of energy due to confinement of electrons in one direction. The density of states for a quantum well is given by

$$g(E) = \frac{4\pi m}{h^2}$$

- Hence, the density of states for a given quantum state will be constant as shown in Fig. 11.3.1(b)(ii). For the different quantum states, it has different constant values as shown in Fig. 11.3.1(b)(iii).
- For a quantum wire, which is a material reduced in two dimensions to nano - scale, there is quantization of energy due to confinement of electrons in two directions. The density of states for a quantum wire is given by

$$g(E) = \frac{2\sqrt{2} m^{1/2}}{h^3} E^{-1/2}$$

- Hence, the density of states for a given quantum state will decrease with increasing energy as shown in Fig. 11.3.1(c)(ii). For the different quantum states, it has variation as shown in Fig. 11.3.1(c)(iii).

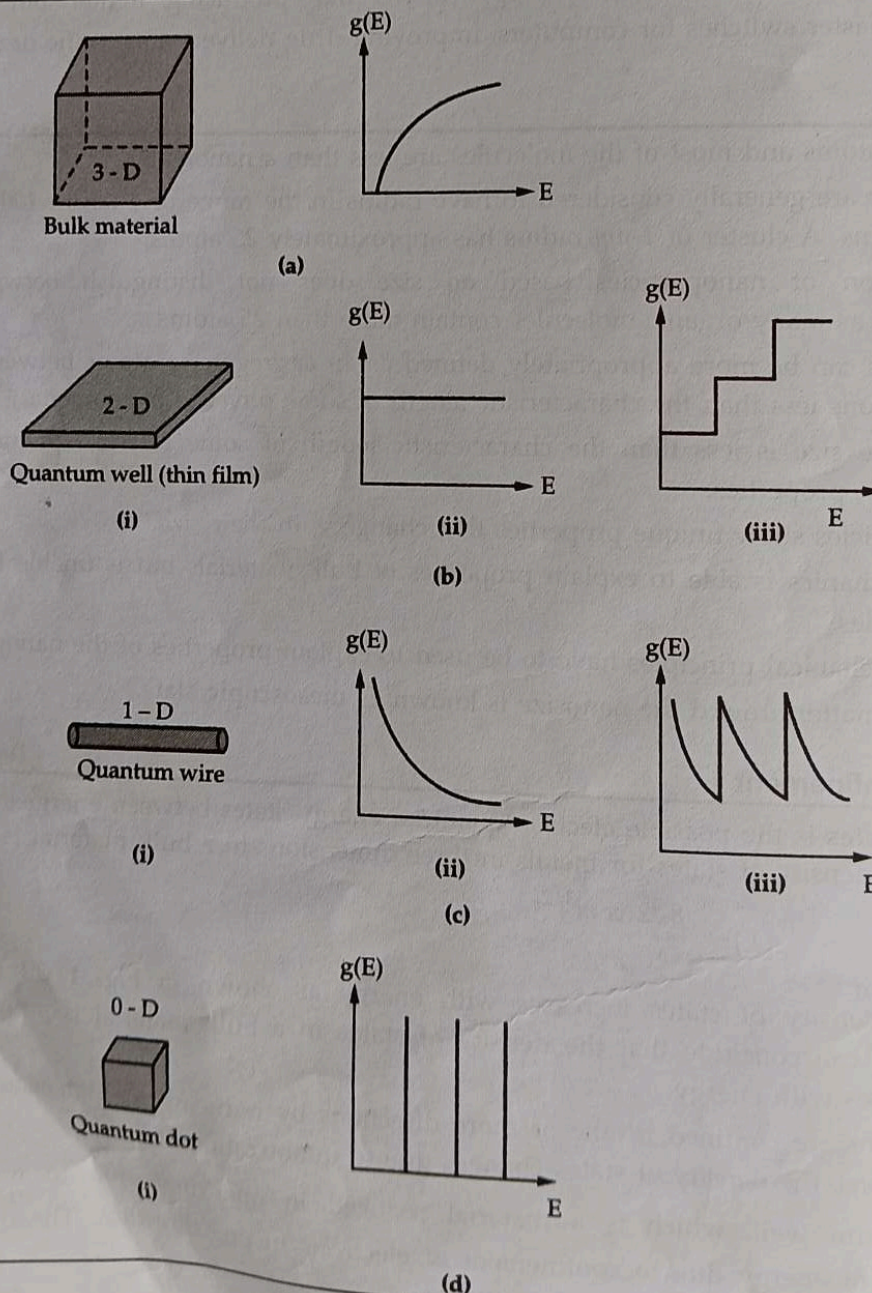


Fig. 11.3.1

- For a quantum dot, which is a material reduced in all three dimensions to nano - scale, there is quantization of energy due to confinement of electrons in all the directions. Hence, for a quantum dot, only some discrete energy states are allowed for the electrons. As only certain energy states are allowed, the density of states has a discrete structure as shown in Fig. 11.3.1(d).

11.4 : Surface to Volume Ratio

- When particle size decreases, the fraction of atoms on the surface to the total number of atoms increases.
- Hence, nanomaterials have larger fraction of atoms at the surface compared to bulk materials.
- Also, the surface energy per unit mass is larger for nanomaterials.
- The surface to volume ratio, which represents the fraction of the atoms on the surface, is larger for nanomaterials.
- The large surface to volume ratio is responsible for higher reactivity of nanomaterials and also their energy band structure.

11.5 : Properties of Nanoparticles

SPPU : Dec.-15, 16, 17, May-16, 17, 18, 19

- The properties of nanoparticles are different from the bulk material.
- The properties of nanoparticles also vary with size and shape.
- Hence different properties can be obtained by changing the size and shape of the nanoparticles.
- Some of the properties of nanoparticles are as follows :

1) Optical properties :

- The colour of nanoparticles is different from the bulk material.
- When a bulk material is reduced in size to a few hundred atoms, the energy band structure of the bulk material changes to a set of discrete energy levels.
- Atomic clusters of different sizes will have different energy level separations.
- As clusters of different sizes have different energy level separations, the colour of the clusters (which are due to transitions between the energy levels) will depend on their size.
- Hence the size of the cluster can be altered to change the colour of a material.
- For example, gold in bulk form appears yellow but gold nanoparticles appear bright red in colour.
- The medieval glass makers produced tinted glass with beautiful variety of colours by dissolving metal particles like gold, silver, cobalt, iron etc.
- Due to these metal nanoparticles, the glasses appear coloured.
- In semiconductor nanoparticles (which are used in quantum dots) there is significant shift in the optical absorption spectra towards blue as the particle size is reduced.

2) Electrical properties :

- The resistivity in bulk matter is mainly due to scattering of electrons by ions and crystal defects.
- In nanostructures, the resistivity mainly depends on scattering from boundaries of nanoparticles when particle size becomes less than the mean free path between collisions.
- Thus smaller particle size increases the resistivity.

- Various types of defects in the lattice also increase the resistivity by limiting the mean free path. But many nanostructures are too small to have internal defects.
- Another effect of reduced size is the confinement of conduction electrons. In bulk conductors, the electrons move freely throughout the entire conductor.
- The situation changes when one or more dimensions of the conductor are made very small.
- In a **quantum well**, the electron will be confined along one dimension but will move freely along the remaining two dimensions.
- If a conducting wire has a long length but very small diameter, the electrons can move freely along the length but will be confined in two mutually perpendicular transverse directions. This configuration is known as a **quantum wire**.
- If all the three dimensions of the conductor are in nanometer range, the configuration is called a **quantum dot** and the electron is confined in all the three dimensions.
- Confinement of electrons to small dimensions leads to quantization of energy.
- The level of doping in semiconductors gives rise to another important phenomenon.
- For typical doping levels of 1 donor impurity atom in 10^8 atoms of semiconductor atoms, a quantum dot of 10^7 semiconductor atoms would have an average of 10^{-1} electrons.
- In other words, on an average, one quantum dot in 10 will have a free electron.
- These result in the phenomena of single-electron tunneling and coulomb blockade.
- The conduction is due to tunneling of electrons through the quantum dot. The electrons are blocked from tunneling except at discrete voltage change positions.
- This phenomenon is called coulomb blockade.
- The I - V characteristic shown in Fig. 11.5.1 is called coulomb staircase.

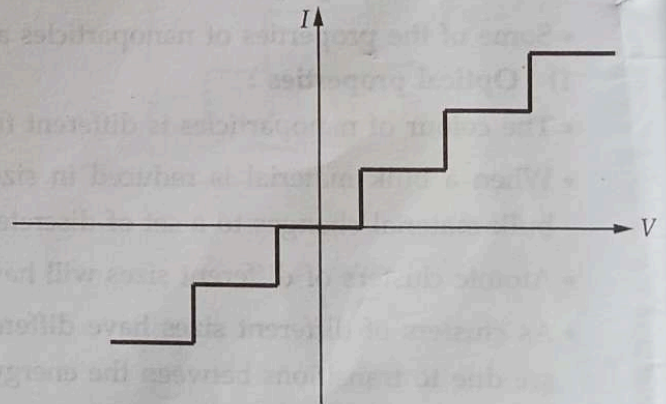


Fig. 11.5.1

3) Mechanical properties :

- Mechanical properties like hardness, elasticity and ductility depend upon the bonds between atoms.
- Imperfections in the crystal structure and impurities result in changes in these properties.
- As the nanoparticles are highly pure and free from imperfections, they show different mechanical properties than the bulk material.
- It has been observed that the Young's modulus decreases in metallic nanocrystals with decrease in particle size.
- The yield stress has been observed to increase with the decrease in grain size in bulk materials with nanosized grains.
- Hence stronger materials can be produced by making materials with nanosized grains.
- The carbon nanotubes are estimated to be about 20 times stronger than steel.

Review Questions

1. Explain any two properties of nano-particles in brief. **SPPU : Dec.-15, Marks 4**
2. Why are the properties of nano-particles different from that of the bulk materials ? Explain any two properties of nano-particles. **SPPU : May-16, Marks 6**
3. Explain the mechanical property of nano-particles. **SPPU : Dec.-16, Marks 2**
4. What is nano technology ? Explain optical and electrical properties of nano-particles. **SPPU : May-17, Marks 6**
5. Explain the Mechanical properties of nanoparticles. **SPPU : Dec.-17, Marks 3**
6. Explain the optical and electrical properties of nanoparticles. **SF SPPU : May-19, Marks 6**

11.6 : Applications of Nanoparticles**SPPU : Dec.-15,16,May-16,17,18,19**

Nanotechnology has found wide ranging applications in many fields. Some of these applications are discussed below.

- 1) **Electronics** : Nanosized electronic components show unique properties which are different from the larger semiconductor components. The semiconductor devices are based on the concept of charge transport only whereas the nanosized components work on the concept of charge as well as spin transport of electrons. This has been used in devices like spin FET, spin LED etc. These devices have increased the data storage capacities of hard disks and have led to small and faster microprocessors.
- 2) **Energy** : Attempts are being made to increase the efficiency of solar cells by using nanotechnology. Another important area of research is the use of hydrogen as a fuel. The main problem with hydrogen is that it is highly combustible and hence cannot be stored easily. Efforts are being made to use carbon nanotubes to trap and store hydrogen.

Nanoparticles are also being used to increase the energy density of rechargeable batteries. Which are used in laptops and mobile phones.

- 3) **Automobiles** : Nanotube composites have better mechanical strength compared to steel but are costly at present. Efforts are being made to develop cheaper nanotube composites that can replace steel which is used to construct the body structure of automobiles. Use of nanoparticles in paints provides thin and smooth coatings.

Nanoparticles are being used to develop light weight and less rubber consuming tyres for automobiles which will increase the mileage of the automobiles. The use of carbon nanotubes for storing hydrogen is being explored so that the automobiles can be run on hydrogen as a fuel.

Nanomaterial catalysts can be used as catalysts to convert the harmful emissions from automobiles to less harmful gases.

- 4) **Space and defence** : Aerogels are porous materials with nanosized pores. They have very low density and are poor conductors of heat. They can be used in spacecrafts, light weight suits and jackets.

Polymer composites using silica fibers and nanoparticles have larger mechanical strength and low temperature coefficient of expansion. They can be used in spacecrafts which have to withstand high temperature and stress conditions during launching and re-entry into the earth's atmosphere.

Satellites and spacecrafts use solar energy.

The efficiency of solar cells can be increased using nanoparticles. The use of nanoparticles will also make the solar cells smaller in size and light weight.

5) **Medical** : Nanoparticles can be used for detection and treatment of cancers and tumours. The nanoparticles are injected into the body and guided towards a specific part. Drugs can be encapsulated in nanocapsules and guided towards any specific part where the drug can be delivered in a controlled manner by opening the capsule at a desired rate using magnetic fields or infrared light. This targeted drug delivery does not affect the healthy organs. Nanotechnology based tests are being developed for fast detection of viruses and antibodies.

6) **Environmental** : Nanoparticle based sensors are capable of detecting water and air pollution due to toxic ions and pesticides with a very high sensitivity. Nanomaterial catalysts can be used as catalysts to convert the harmful emissions from industries and automobiles to less harmful gases.

Nanotubes can be used to store hydrogen fuel which, when used in automobiles, will reduce harmful emissions.

7) **Textiles** : The use of nanotechnology in textile industry has led to the development of water repellent and wrinkle free clothes.

8) **Cosmetics** : Zinc oxide and titanium oxide nanoparticles are used in sunscreen lotions which protect the skin from the ultraviolet radiations. These nanoparticles absorb ultraviolet radiations. Nanoparticle based dyes and colours are harmless to the skin and hence are used in hair creams, gels and hair dyes.

Review Questions

1. Explain the applications of nano-particles in electronic industry.
2. Explain the applications of nano-particles in the field of automobiles.
3. Explain the application of nano particles in medical and automobile field.
4. State applications of nano-particles. Explain any one application.
5. List the applications of nanotechnology in the field of automobile. Explain any one application in brief.

SPPU : Dec.-15, Marks 6

SPPU : May-16, Marks 3

SPPU : Dec.-16, Marks 3

SPPU : May-17,19, Marks 3

SPPU : May-18, Marks 3